MILITARY STANDARD

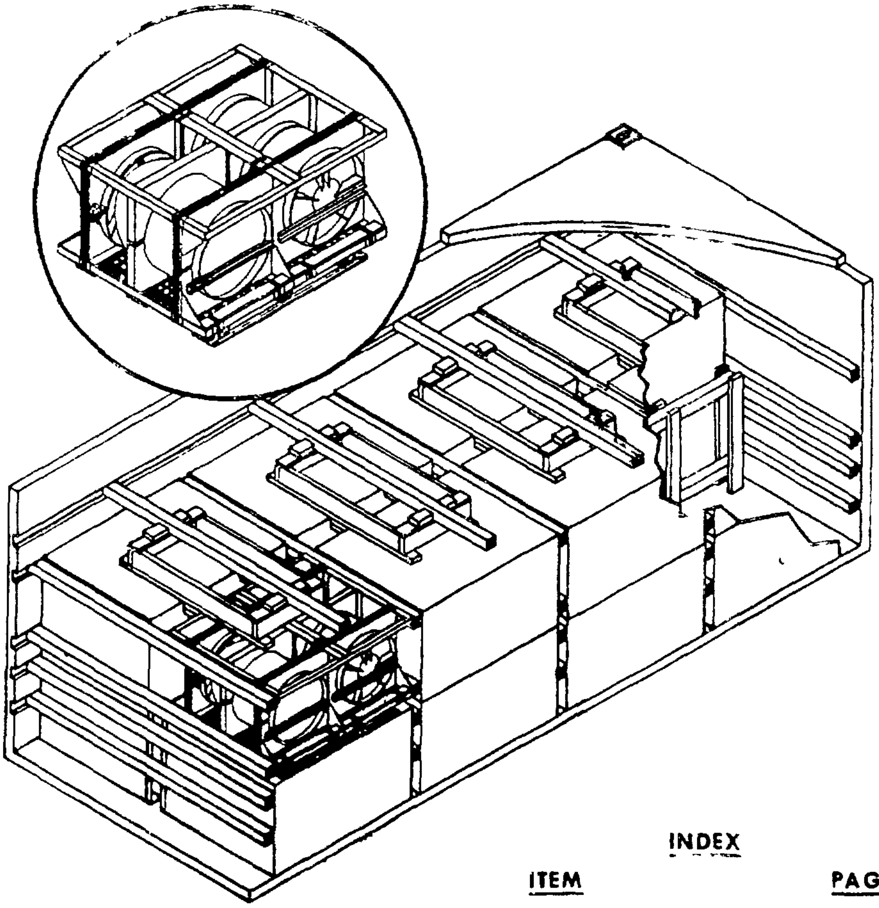
MIL-STD-1386-57
(NAVY)
9 JUNE 1975

LOADING OF HAZARDOUS MATERIALS AND ASSOCIATED ITEMS IN MILVAN CONTAINERS

EXPLOSIVE SECTION MK 2 MOD 2
IN CRATE MK 103 MOD 0
FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DWG	WR 54/200
WEIGHT OF UNIT LOAD	1374 LBS
DIMENSIONS	52 L x 42 W x 30 1/4 H
CUBE	38 2 CU FT
DOT HAZARD CLASSIFICATION	EXPLOSIVES A
COAST GUARD CLASS	XA



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NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CARLOADING MILITARY STANDARDS, REFER TO THE INDEX TO STANDARDS, MIL-HDBK-236 (NAVY)

FSC 8140

AUTHORIZED FOR TRAILER-ON-FLAT-CAR (TOFC), CONTAINER-ON-FLAT-CAR (COFC), CONTAINERSHIP AND HIGHWAY SHIPMENTS

5/9/75
SIGNATURE TECHNICAL DIRECTION [illegible] DATE
A P Kolb 5/13/75

APPROVED BY US COAST GUARD

N B Chung CAPT USCG
2-27-75

APPROVED BY BUREAU OF EXPLOSIVES

8/12/75

ORIGINATOR
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY

GENERAL NOTES

1 THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR LOADING THE MILVAN CONTAINER WITH FLEET ISSUE UNIT LOAD OF MK 2 MOD 2 EXPLOSIVE SECTIONS FOR MK 57 UNDERWATER MINES IN CRATE MK 103 MOD 0 THE SECTIONS SHALL BE PALLETIZED IN ACCORDANCE WITH WR-54/200

2 THE MILVAN SHALL BE EQUIPPED WITH A MECHANICAL LOAD-BRACING SYSTEM THAT MEETS THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C

3 IN ADDITION TO THE BELT RAILS REQUIRED BY PAMPHLET 6C, THE MILVAN CONTAINER HAS BELT RAILS AT THE 72 AND 83 INCH HEIGHTS FROM THE CONTAINER FLOOR

CAUTION

WHEN POSITIONING CROSSMEMBERS MAKE A VISUAL CHECK TO ENSURE THAT THE CROSSMEMBERS ARE LOCKED SECURELY IN PLACE

4 CONTAINERS ARE EQUIPPED WITH 25 CROSSMEMBERS UNUSED CROSSMEMBERS MUST REMAIN WITH THE CONTAINER AND SHALL BE FASTENED INTO BELT RAILS FOR SHIPMENT

5 WHEN THE LONGITUDINAL SLACK BETWEEN A CROSSMEMBER AND THE UNIT LOAD IS GREATER THAN ONE INCH, USE FILL MATERIAL TO SHIM BETWEEN THE CROSSMEMBERS AND THE UNIT LOAD TO FILL THE VOID TWIST TIE WITH 3 PIECES OF 16 GAUGE SOFT ANNEALED WIRE, AS SHOWN IN DETAIL D THIS SHIM MAY BE ADDED OR OMITTED AS REQUIRED

6 APPLICABLE MATERIAL SPECIFICATIONS

DUNNAGE LUMBER - FED SPEC MM-L-751

NAILS - FED SPEC FF-N-105, TYPE II, STYLE 10, COMMON BRIGHT

7 AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS INSIDE THE CONTAINER IN AN ACCESSIBLE AREA, CLOSE AND SEAL CONTAINER DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF THE CONTAINER FOR THE APPROPRIATE MODE OF TRANSPORTATION AS PRESCRIBED IN OP 2165 VOL I

TOFC/COFC,CONTAINERSHIP AND HIGHWAY

1 THE CONTAINER LOAD CONSISTS OF 16 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS DOCUMENT

2 THE LOADED MILVAN CONTAINER SHALL BE ON A 20 FT MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES WHEN BEING MOVED IN TOFC SERVICE

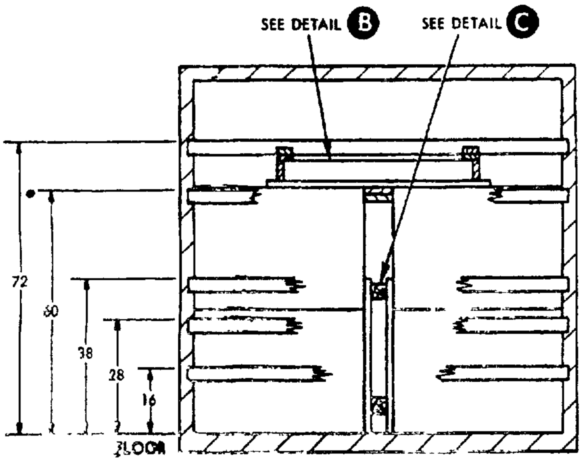
3 WHEN THE LOADED CONTAINER IS IN TOFC OR COFC SERVICE, IT SHALL BE ON FLAT RAILCARS THAT COMPLY WITH BUREAU OF EXPLOSIVES PAMPHLET 6C

4 THE LOADED MILVAN MAY BE TRANSPORTED OVER PUBLIC HIGHWAYS ON A MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES OR ON SUITABLE COMMERCIAL EQUIPMENT OF APPROPRIATE CAPACITY AND MEETS THE REQUIREMENTS OF THE CODE OF FEDERAL REGULATIONS

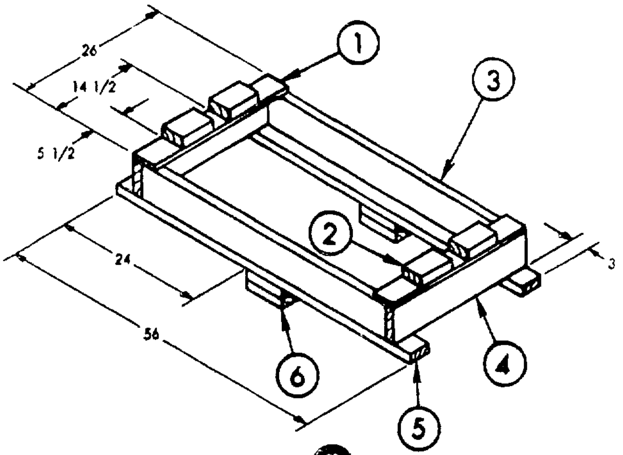
NOTE

IN CERTAIN STATES THE GROSS VEHICLE WEIGHT (G V W) IS LIMITED BY THE DISTANCE BETWEEN THE AXLES OF THE ASSEMBLED RIG CARE SHOULD BE TAKEN TO SEE THAT THE PROPOSED G V W DOES NOT EXCEED THESE LIMITATIONS

5 SHIM, PIECE 1, OF HOLD DOWN ASSEMBLY, DETAIL B, MAY CONSIST OF MORE THAN ONE PIECE OF DIFFERENT THICKNESSES AND SHOULD CAUSE THE ASSEMBLY TO FILL THE VOID BETWEEN THE LADING AND THE BOTTOM OF THE CROSSMEMBER VOIDS BETWEEN HOLD DOWN CROSSMEMBERS AND SHIM SHALL NOT EXCEED ONE HALF INCH CENTER ASSEMBLY OVER UNIT LOADS

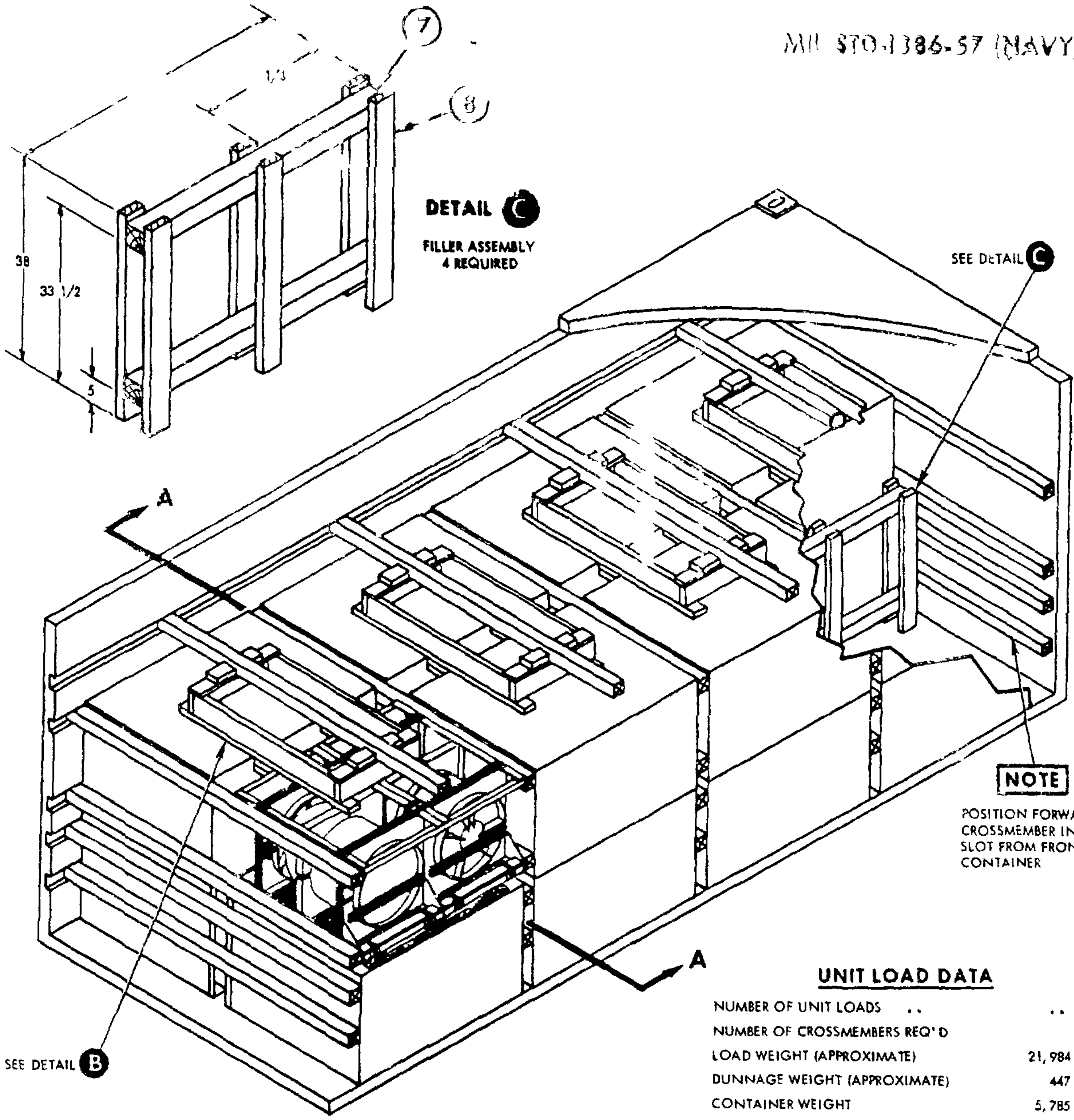


SECTION A-A



DETAIL B

HOLD DOWN SHIM ASSEMBLY
4 REQUIRED

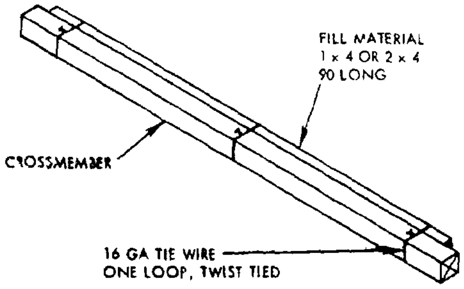


NOTE

POSITION FORWARD CROSSMEMBER IN 1ST SLOT FROM FRONT OF CONTAINER

UNIT LOAD DATA

NUMBER OF UNIT LOADS	16
NUMBER OF CROSSMEMBERS REQ'D	24
LOAD WEIGHT (APPROXIMATE)	21,984 LBS
DUNNAGE WEIGHT (APPROXIMATE)	447 LBS
CONTAINER WEIGHT	5,785 LBS
GROSS WEIGHT (APPROXIMATE)	28,216 LBS



DETAIL D

SEE GENERAL NOTE 5

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
8	VERTICAL	2 x 4 x 38	24	7	3/JOINT	12d
7	HORIZONTAL	4 x 4 x 52	8	SEE 8	-	-
6	SPACER	2 x 4 x TO SUIT	16	5, 6	3	16d
5	LOWER BEARING PIECE	2 x 4 x 56	8	4 3	3 2/JOINT	16d 16d
4	RISER	2 x 6 x 26	8	3	3/JOINT	10d
3	SUPPORT	2 x 6 x 47	8	SEE 4 & 5	-	-
2	RETAINER	2 x 4 x 6	16	1	2	10d
1	SHIM	4 WIDE x 26 LONG x THICKNESS TO SUIT	AS REQ'D	3 & 4	3	10d

LIST OF MATERIALS AND NAILING DATA

TOFC/COFC, CONTAINERSHIP AND HIGHWAY

REVIEW ACTIVITY
NAVY - OS

PREPARING ACTIVITY
NAVY - OS
(PROJECT NO 8140-N282)

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION
MIL-STD-1386-57 (NAVY)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JUN 68 1426

REPLACES EDITION OF 1 OCT [illegible] WHICH MAY BE USED

S/N-0102-014-1801 C-25264

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POSTAGE AND FEES PAID

OFFICIAL BUSINESS

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Naval Weapons Handling Laboratory (803)
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